

Spatial IMA1 regulation restricts root iron acquisition on MAMP perception

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Iron is critical during host–microorganism interactions^{1–4}. Restriction of available iron by the host during infection is an important defence strategy, described as nutritional immunity⁵. However, this poses a conundrum for externally facing, absorptive tissues such as the gut epithelium or the plant root epidermis that generate environments that favour iron bioavailability. For example, plant roots acquire iron mostly from the soil and, when iron deficient, increase iron availability through mechanisms that include rhizosphere acidification and secretion of iron chelators^{6–9}. Yet, the elevated iron bioavailability would also be beneficial for the growth of bacteria that threaten plant health. Here we report that microorganism-associated molecular patterns such as flagellin lead to suppression of root iron acquisition through a localized degradation of the systemic iron-deficiency signalling peptide Iron Man 1 (IMA1) in *Arabidopsis thaliana*. This response is also elicited when bacteria enter root tissues, but not when they dwell on the outer root surface. IMA1 itself has a role in modulating immunity in root and shoot, affecting the levels of root colonization and the resistance to a bacterial foliar pathogen. Our findings reveal an adaptive molecular mechanism of nutritional immunity that affects iron bioavailability and uptake, as well as immune responses.

Iron is an essential nutrient for organismal growth throughout all branches of life^{10,11}. Although iron is among the most common elements on our planet, its bioavailability in most environments is low and is a limiting factor for growth. Consequently, strong competition for iron is common between organisms. In mammals, iron levels have a direct impact on the composition of the gut microbiota^{1,2} and can modulate host inflammatory responses restricting iron availability for pathogenic microorganisms⁵. In plants, iron-limited soil environments trigger the activation of coumarin secretion in roots, which contributes to the alteration of the root-associated microbiota^{3,4}. Conversely, the soil-borne bacterial community can also have a considerable impact on root iron acquisition^{3,12,13}. On the other side of the spectrum, pathogens can compete for and restrict iron for plants^{14,15}, plant immunity responses affect bacterial iron homeostasis¹⁶ and beneficial microorganisms can produce siderophores that pathogens cannot use, consequently suppressing pathogen growth^{17,18}. Overall, iron has an important but complex role in regulating plant–microorganism interactions. However, much about this multifaceted interaction that includes plants, beneficial or commensal and pathogenic microorganisms in the rhizosphere¹⁹ remains to be learned.

Flg22 represses root responses to –Fe

We previously found a pronounced interplay of responses to low iron and immunity within the first few hours after roots were exposed to

an iron-depletion environment²⁰. To investigate this interplay at a longer timescale, we grew *Arabidopsis* seedlings in iron-sufficient and low-iron media with and without the elicitor flg22, a peptide fragment of bacterial flagellin (Fig. 1a). Compared with seedlings grown under iron-sufficient conditions (+Fe), seedlings grown under low-iron conditions (–Fe) were slightly more chlorotic and contained less iron. This was exacerbated substantially after co-treatment with flg22 (Fig. 1a–c and Extended Data Fig. 1a), but not when treated with a non-immunogenic form of flagellin, flg20 (Extended Data Fig. 1b,c). We therefore hypothesized that flg22 abolishes root iron uptake. To test this hypothesis, we measured ferric chelate reductase (FCR) activity and expression of IRON REGULATED TRANSPORTER1 (IRT1), which are part of the canonical iron-deficiency response in *Arabidopsis* and facilitate iron uptake. Whereas FCR activity in roots was induced under –Fe without flg22 treatment, FCR activity was not induced under –Fe conditions in the presence of flg22 (Fig. 1d and Extended Data Fig. 1d). Similarly, a lack of iron triggered the induction of the *pIRT1::NLS-2xYpet* marker²⁰ in the epidermal cells of the differentiation zone, and treatment with flg22, but not flg20, abolished these responses (Fig. 1e,f and Extended Data Fig. 1e). IRT1 protein levels reflected this response (Fig. 1g and Extended Data Fig. 1f–h). This was dependent on flg22 reception by the FLS2 immune receptor as flg22-induced chlorosis, and flg22-triggered FCR activity and IRT1 repression under –Fe conditions were abolished in *fls2* mutants (Extended Data Fig. 1i–m). Overall, these data demonstrated

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